8 Discrete Mathematics: Combinatory (12)

8.1 Circular Permutation (1)

8.1.1 Circular Permutation: NIELIT 2017 DEC Scientist B - Section B: 15

In how many ways 8 girls and 8 boys can sit around a circular table so that no two boys sit together?

A. $(7!)^2$ B. $(8!)^2$ C. $7!8!$ D. $15!$

nielit2017dec-scientistb discrete-mathematics circular-permutation

Answer key☟

8.2 Combinatory (3)

8.2.1 Combinatory: NIELIT 2016 DEC Scientist B (IT) - Section B: 34

Mala has a colouring book in which each English letter is drawn two times. She wants to paint each of these 52 prints with one of k colours, such that the colour-pairs used to colour ay two letters are different. Both prints of a letter can also be coloured with the same colour. What is the minimum value of k that satisfies this requirement?

A. 9 B. 8 C. 7 D. 6

nielit2016dec-scientistb-it discrete-mathematics combinatory

Answer key☟

8.2.2 Combinatory: NIELIT 2016 MAR Scientist B - Section B: 2

The number of ways in which a team of eleven players can be selected from 22 players including 2 of them and excluding 4 of them is

A. ${}^{16}C_{11}$ B. ${}^{16}C_5$ C. ${}^{16}C_9$ D. ${}^{20}C_9$

nielit2016mar-scientistb discrete-mathematics combinatory

Answer key☟

8.2.3 Combinatory: NIELIT 2017 July Scientist B (IT) - Section B: 14

There are four bus lines between A and B; and three bus lines between B and C. The number of way a person roundtrip by bus from A to C by way of B will be

A. 12 B. 7 C. 144 D. 264

nielit2017july-scientistb-it discrete-mathematics combinatory

Answer key☟

8.3 Counting (1)

8.3.1 Counting: NIELIT Scientist B 2020 November: 21

There are 6 boxes numbered $1, 2, \ldots \ldots, 6$. Each box is to be filled up either with a red or a green ball in such a way that at least 1 box contains a green ball and the boxes containing green balls are consecutively numbered. The total number of ways in which this can be done is :

A. 18 B. 19 C. 20 D. 21

nielit-scb-2020 counting combinatory

Answer key☟

8.4 Permutation and Combination (2)

8.4.1 Permutation and Combination: NIELIT 2021 Dec Scientist A - Section A: 37

How many eight letter words can be formed from the letters of the word "COURTESY" beginning with C and ending with Y ?

A. 120 B. 256 C. 720 D. 750

nielit2021dec-scientista counting permutation-and-combination

Answer key☟

8.4.2 Permutation and Combination: NIELIT Scientist B 2020 November: 118

The number of 4 digit numbers which contain not more than two different digits is:

A. 576 B. 567 C. 513 D. 504

nielit-scb-2020 combinatory counting permutation-and-combination

8.5 Recurrence Relation (4)

8.5.1 Recurrence Relation: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 6

The solution of the recurrence relation

$a_r = a_{r-1} + 2a_{r-2}$ with $a_0 = 2, a_1 = 7$ is

A. $a_r = (3)^r + (1)^r$
B. $2a_r = (2)^r/3 – (1)^r$
C. $a_r = 3^{r+1} – (-1)^r$
D. $a_r = 3(2)^r – (-1)^r$

nielit2017oct-assistanta-cs discrete-mathematics combinatory recurrence-relation

Answer key☟

8.5.2 Recurrence Relation: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 34

The solution of the recurrence relation

$a_r = a_{r-1} + 2a_{r-2}$ with $a_0 = 2, a_1 = 7$ is

A. $a_r = (3)^r + (1)^r$
B. $2a_r = (2)^r/3 – (1)^r$

C. $a_r = 3^{r+1} - (-1)^r$ D. $a_r = 3(2)^r - (-1)^r$

nielit2017oct-assistanta-it combinatory recurrence-relation

Answer key☟

8.5.3 Recurrence Relation: NIELIT 2022 April Scientist B | Section B | Question: 48

The particular solution of the recurrence relation $a_{r+2} - 4a_{r+1} + 4a_r = 2^r$ is:

A. $r.2^r$ B. $r(r-1)2^{r-1}$ C. $r(r-1)2^{r-2}$ D. $r(r-1)2^{r-3}$

nielit2022apr-scientistb combinatory recurrence-relation

Answer key☟

8.5.4 Recurrence Relation: NIELIT Scientific Assistant A 2020 November: 51

Given $r_{12} = 0.6, r_{13} = 0.5$ and $r_{23} = 0.8,$ the value of $r_{12.3}$ is :

A. 0.47 B. 0.40 C. 0.74 D. 0.64

nielit-sta-2020 combinatory recurrence-relation

8.6 Summation (1)

8.6.1 Summation: NIELIT 2021 Dec Scientist A - Section B: 59

Let $\mathrm{C}(n,\ r) = \binom{n}{r}$. The value of $\sum_{k=0}^{20}(2k+1)\mathrm{C}(41, 2k+1)$, is :

A. $40(2)^{40}$ B. $40(2)^{39}$ C. $41(2)^{40}$ D. $41(2)^{39}$

nielit2021dec-scientista combinatory permutation-and-combination summation

Answer key☟

Answer Keys

8.1.1	C	8.2.1	C	8.2.2	C	8.2.3	C	8.3.1	D
8.4.1	C	8.4.2	A	8.5.1	D	8.5.2	D	8.5.3	D
8.5.4	B	8.6.1	D						